Eight bounded rank-four hyperbolic quotient searches

Independent recovery of fixed-source Table 3

Abstract

We certify finite marked quotients for eight diagrams appearing in Table 3 of Fernandes–Leemans–Weiss. Complete bounded normal-subgroup searches, followed by intersection and actual coset-incidence tests, give 59 positive marked kernels. We distinguish universal spherical residues from residues whose diagrams are finite. In particular no regular hypertope with either local convention realizes the printed order 336 for $(3,4,3,4)$ or order 2160 for $(3,5,3,4)$. Exact named extension descriptions and the ninth diagram $(3,5,3,5)$ are outside this component.

1 Conventions and the finite assertion

Write $I = \{1,2,3,4\}$. For a symmetric Coxeter matrix M, put

$$W(M) = \langle s_i \ (i \in I) \mid s_i^2 = 1, (s_i s_j)^{M_{ij}} = 1 \ (i < j) \rangle.$$

In a square (p,q,r,s) the edges $12, 23, 34, 41$ have those labels, and the diagonals have label 2. In a string $[p,q,r]$ the edges are $12, 23, 34$. For the fork $[5,3_3]$ the edges $12, 13, 14$ have labels $5, 3, 3$. Unlisted off-diagonal entries are 2. These conventions agree, up to relabelling, with the native diagrams on source physical page 12 and the types on physical page 14.

For a quotient $H = \langle t_1, t_2, t_3, t_4 \rangle$, let $H_J = \langle t_j : j \in J \rangle$ and $P_i = H_{I \setminus \{i\}}$. The *broad* column below requires a regular hypertope with this exact diagram; its rank-three residues have finite diagrams. The *universal* column additionally requires each P_i to be the full finite Coxeter group for the corresponding rank-three subdiagram. This distinction matters: the fixed source defines spherical by a finite diagram on page 5, but explicitly restricts to spherical, rather than projective, local geometries in its euclidean discussion on page 10. We do not silently resolve that terminological tension for Section 6.

Diagram	Bound	Kernels	Positive	Broad minimum	Universal minimum
$(3,3,3,4)$	7680	14	10	168	1920
$(3,3,3,5)$	7200	5	2	2520	7200
$(3,4,3,4)$	9216	45	21	168	1152
$(3,5,3,4)$	2160	8	1	1080	> 2160
$[3,5,3]$	8160	5	2	660	8160
$[4,3,5]$	7680	17	7	1920	3840
$[5,3,5]$	8160	6	3	3420	7200
$[5,3_3]$	7680	20	13	960	3840

Here kernels include the whole group (index 1). Positive means distinct normal kernels for the fixed ordered presentation, not isomorphism classes after diagram automorphisms. There are 120 kernels in total and 59 positives. The entry > 2160 is only an exclusion through that bound, not an existence or minimum assertion beyond it. All displayed attained minima have explicit permutation witnesses. The original source's group names are not inferred from orders or from descriptive strings printed by GAP.

2 A complete coset test

Suppose the involutions t_i generate a finite group H, have exact pair orders M_{ij}, and satisfy

$$H_A \cap H_B = H_{A\cap B} \qquad (A, B \subseteq I). \tag{1}$$

Form the incidence system whose objects of type i are right cosets $P_i g$, with incidence given by nonempty intersection. For $J \subseteq I \setminus \{k\}$ let $E(J,k)$ be the set of type-k objects incident with every base object P_i, $i \in J$. The following exact test is applied for all 32 choices:

$$E(J,k) = \{P_k h : h \in H_{I\setminus J}\}. \tag{2}$$

The GAP implementation compares the actual sets in the right-coset permutation action. The independent integer-permutation implementation constructs every right coset as a literal set and checks the equivalent cardinality $|E(J,k)| = |H_{I\setminus J}|/|H_{I\setminus(J\cup\{k\})}|$. The orbit on the right of (2) is always contained in the left, and its stabilizer is the denominator subgroup by (1); consequently this cardinality check really implies set equality.

We give the argument that this tests flag-transitivity, including flags which might otherwise be omitted by a chamber-only model. Induct on the size of a flag. Normalize its first $|J|$ objects to the base flag using the induction hypothesis. Any further incident type-k object lies in $E(J,k)$ and hence can be normalized by the base-flag stabilizer $\bigcap_{i\in J} P_i = H_{I\setminus J}$. Thus every flag is an image of a base flag. In particular every flag extends to a chamber, and H is flag-transitive on the actual incidence system.

The chamber stabilizer is trivial by (1). A corank-one flag has exactly two completions because its stabilizer is the two-element group generated by its remaining mark. More generally, the stabilizer of a base flag is generated by the remaining marks. Their chamber-adjacency graph is connected. For a residue of rank at least two these successive adjacent chambers share an object, so its incidence graph is connected. The system is therefore a thin, residually connected geometry. Finally any type-preserving automorphism fixing a chamber fixes its unique adjacent chamber of each type and then every chamber. It is the identity. Consequently the full type-preserving group is precisely H, and the geometry is a regular hypertope.

All maximal subdiagrams here are finite. Their universal orders, in our chosen vertex order, are respectively

$$\begin{array}{ll} 3334 : (24,48,48,24), & 3335 : (24,120,120,24), \\ 3434 : (48,48,48,48), & 3534 : (120,48,48,120), \\ 353 : (120,12,12,120), & 435 : (120,20,16,48), \\ 535 : (120,20,20,120), & 5333 : (8,24,120,120). \end{array}$$

These are the orders of A_3, B_3, H_3 or products of rank-one and dihedral Coxeter groups. Equality with these orders makes the natural surjection of each finite local Coxeter group injective. Otherwise the actual local orders are retained explicitly in the witness records; the broad column does not assume an injective local quotient.

3 Exhaustion, source corrections, and retained premises

Every finite regular hypertope with the prescribed diagram supplies its distinguished involutions, hence an epimorphism $W(M) \to H$. Its kernel is normal and has index $|H|$. This implication needs no injection on local parabolics. Relabelling a source-equivalent diagram gives a marking of our fixed presentation, so does not evade the normal-kernel search. We enumerate all normal subgroups

through the displayed bound, discard quotients with wrong mark or pair orders, and apply (1) and (2). The only index-only prefilter is divisibility by $\mathrm{lcm}_{i<j}(2M_{ij})$: each genuine rank-two marked subgroup has order $2M_{ij}$. Thus no required quotient is lost.

The enumeration uses GAP 4.15.1 and LINS 0.9, `LowIndexNormalSubgroupsSearchForAll`, with the unmodified default options. Its documented contract is all normal subgroups through the index bound, not one quotient per abstract isomorphism type. The installed limit is 10^7, checked before each search. The default has no early termination or branch cut, and uses `UseLIS=false`. The algorithm and its exact computer implementation are retained computational premises. In particular LINS's tables of characteristically simple groups, Schur multiplier primes, and transitive socle-overgroups are classification inputs; we do not claim to reprove or regenerate their completeness. The local documentation attributes the algorithm to Firth's 2005 thesis and explains its pregenerated tables. Exact executable/table bytes and clean terminal receipts are bound in the packet. This is not a formal verification of GAP or LINS.

There are exactly three kernels of index 336 for the first disputed square and one of index 2160 for the second. Every resulting marking fails (1), even allowing smaller finite-diagram local residues. Their compact tuples are retained and independently checked by literal permutation closure. The source order 336 and 2160 rows therefore do not describe regular hypertopes of their printed diagrams. A further negative control is the order 360 quotient of $(3, 5, 3, 4)$: it satisfies all 256 intersections but has four, rather than two, type1 completions of a particular rank-three base flag. It illustrates why local orders or the intersection property alone cannot replace the incidence test.

There is also an elementary, more restricted obstruction at 336: with four universal B_3 residues, choose two maximal parabolics whose remaining edge has label3. Their intersection would have order 6, and their product set would have $48^2/6 = 384$ elements. This argument alone does *not* exclude projective local residues; the unqualified index 336 exclusion above uses all three kernels.

Fresh and retained positive inventories are compared by a bijection of ordered markings: an explicitly checked isomorphism sends each of the four old marks to the corresponding new mark. This is equivalent to equality of their kernels in the fixed presentation. Every record is matched once, so multiplicities are preserved; equality of compact permutation bytes is neither required nor presumed. The literal JSON inputs are also compared with the exact GAP records before their independent audit.

The positive checks are repeated using independent standard-library permutation closure and explicit coset sets. Neither compacting the GAP actions nor printing structural descriptions supplies a completeness premise. This component proves the finite assertions displayed above; it does not finish Table 3's exact named extensions, the ninth square, rank-five existence, the source's other classifications, or the whole original. Initial universal-only checks and all capped square 3535 attempts are preserved as history and are not substituted for the broad complete executions.

Attribution. The target rows, diagrams and minimum claims belong to Fernandes–Leemans–Weiss, *An Exploration of Locally Spherical Regular Hypertopes*, fixed 16-page source, Sections 2 and 6, Tables 2–3. The parent supplied the first local-order square 3434 probe and identified the source's local-scope ambiguity. The present author supplied the full incidence/intersection implementation, independent literal audit and bounded core synthesis. A separate reader inspected the LINS documentation and table boundary. None of these activities asserts priority for the underlying finite-group algorithms or adds a new measurement or question.